Pro Football Sports Science Research Update

June 2024

A little pepper-upper? Systematic review of randomized controlled studies on capsaicinoids, capsinoids, and exercise performance. International Journal of Sport Nutrition and Exercise Metabolism, 33(5): 291-301, 2023.

A randomized trial of healthy weight gain in athletic individuals. Medicine and Science in Sports and Exercise, Ahead of Print: 1-55, 2024.

Acute:chronic workload ratio of professional soccer players preceding hamstring muscle injuries: A 2-season retrospective study. Sport Sciences for Health, Ahead of Print: 1-7, 2023.

Analysis of concussions reported in American professional football injury reports in the 2019 through 2023 regular season. World Neurosurgery, Ahead of Print: 1-23, 2024.

Biceps femoris muscle-tendon strain during an entire overground sprint acceleration: A biomechanical explanation for hamstring injuries in the acceleration phase. Sports Biomechanics, Ahead of Print: 1-15, 2024.

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Does a broad-spectrum cannabidiol supplement improve performance in a 10-min cycle ergometer performance test? European Journal of Sport Science, Ahead of Print: 1-8, 2024.

Does it matter? Isometric or isokinetic assessment of quadriceps strength symmetry 9 months after ACLR in collegiate athletes. Sports Health, Ahead of Print: 1-9, 2024.

Doping prevalence among U.S. elite athletes subject to drug testing under the World Anti-Doping Code. Sports Medicine – Open, 10:57, 2024.

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Effect of personalized sodium replacement on fluid and sodium balance and thermophysiological strain during and after ultraendurance running in the heat. International Journal of Sports Physiology and Performance, 19(2): 105-115, 2023.

Effectiveness of recovery strategies after training and competition in endurance athletes: An umbrella review. Sports Medicine – Open, 10:55, 2024.

Effects of chronic static stretching interventions on jumping and sprinting performance: A systematic review with multilevel meta-analysis. Frontiers in Physiology, 15:1372689, 2024.

Epidemiology of injury and illness in North American professional men's soccer: Comparing COVID-19 lockdown with previous season. Sports Health, Ahead of Print: 1-7, 2024.

Evaluating a 14-week neck strengthening protocol for neuromuscular indicators associated with head and neck trauma. Sports Medicine and Health Science, Ahead of Print: 1-7, 2024.

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A little pepper-upper? Systematic review of randomized controlled studies on capsaicinoids, capsinoids, and exercise performance. International Journal of Sport Nutrition and Exercise Metabolism, 33(5): 291-301, 2023.

Abstract - Capsaicinoids and capsinoids are bioactive compounds mostly found in peppers. Although preclinical studies have reported that these compounds can improve exercise performance due to transient receptor potential vanilloid subtype 1 (TRPV1)-mediated thermogenesis, sympathetic modulation, and releasing calcium, it is still unclear how they affect exercise performance in humans as ergogenic supplements. Conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses reporting guide 2020, this systematic review examined the ergogenic effect of capsaicinoids and capsinoids on exercise performance in healthy adults. A total of 19 randomized placebo-controlled trials were included in the study. Studies were accessed by searching five databases (PubMed, Scopus, SPORTDiscus, Web of Science, and Cochrane Library). The quality of the studies was evaluated using the Cochrane risk-of-bias assessment tool. According to the study results, 10 studies examining the effect of capsaicinoid and capsinoid supplements on exercise performance reported positive effects. Also, the effect of capsaicinoids and capsinoids on exercise performance is more pronounced in resistance training. This difference, which varies according to the type of exercise, may be due to the correlation between capsaicin transient receptor potential vanilloid subtype 1 and insulin-like growth factor-1.

A randomized trial of healthy weight gain in athletic individuals. Medicine and Science in Sports and Exercise, Ahead of Print: 1-55, 2024.

Overview: Many athletes desire weight gain, ideally as lean body mass (LBM), to improve athletic performance. These athletes are commonly encouraged to increase energy intake by ~500 kcal/day with emphasis on adequate protein, carbohydrate, and judicious inclusion of healthy fat containing energy-dense foods, such as peanuts/peanut butter, along with rigorous resistance training (RT). These guidelines target gains of ~0.23 kg/week. However, little is known about the efficacy of such regimens.

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Purpose: To evaluate the outcomes of a 10-week diet and exercise regimen designed to promote healthy weight gain with excess energy from peanut containing or high-carbohydrate foods. Methods: 19 male and 13 female athletes were randomly assigned to receive 500 additional kcal/day above typical intake through provision of either peanut-based whole foods/snacks (PNT group) or a similar, high-carbohydrate, peanut-free snack (CHO group) along with supervised, whole-body RT (3 days/week for 60-120 minutes). Body composition was assessed by dual-energy x-ray absorptiometry at baseline and post-intervention.

Results: Total body mass (TBM) increased 2.2±1.3 kg with 1.5±1.1 kg as LBM after week 10. The PNT group (N=16; 27±7 years; 10 men, 6 women) gained less TBM than the CHO group (N=16; 23±3 years; 9 men, 7 women) (1.6±1.1 kg vs 2.7±1.2 kg, respectively, P=0.007) with no differences in LBM (1.2±1.1 kg vs 1.9±1.0 kg, P=0.136).

Conclusions: These results suggest that the addition of 500 kcal/day from whole foods/snacks in combination with a rigorous RT program promotes a similar weight gain of ~0.22 kg/week, primarily as LBM, over 10 weeks in both male and female athletes. However, snack macronutrient content may impact the effectiveness of this regimen.

Acute:chronic workload ratio of professional soccer players preceding hamstring muscle injuries: A 2-season retrospective study. Sport Sciences for Health, Ahead of Print: 1-7, 2023.

Purpose: Given the controversial use of the Acute:Chronic Workload Ratio (ACWR) in determining the availability of soccer players for training and matches, this study aimed at verifying how many professional soccer players had an ACWR above 1.5 (considered a 'danger zone') immediately before the exposure (training or match) in which they sustained a hamstring muscle injury.

Methods: Forty-eight male soccer players of a professional team were monitored through global positioning system (GPS) in this 2-season retrospective study (10 participated in both seasons and 38 in a single season). The ACWR of total distance, high-speed running distance, very high-speed

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running distance, and the number of actions performed>19 km/h were calculated before all hamstring injuries and included in the analysis.

Results: Twenty players sustained hamstring injuries. There was no time effect on the group's daily workload or the average workload over the 4 weeks before the injury (p>0.05). Fifteen players (i.e., 75%) had ACWRbelow1.5 for the four GPS metrics immediately before the training or match in which they sustained a hamstring injury. Five players (25%) were classified in the 'danger zone' (i.e., ACWR>1.5) for the very high-speed running distance; 3 of these players were also in the 'danger zone' for high-speed running distance and the number of actions>19 km/h, and 2 players for the total distance.

Conclusion: Since three-quarters of the hamstring injuries affecting professional soccer players were preceded by ACWR values below 1.5 in all GPS metrics, it is prudent for medical/coaching staff exercise caution when considering the ACWR 'danger zone' as the primary metric for determining player availability for training or matches.

Analysis of concussions reported in American professional football injury reports in the 2019 through 2023 regular season. World Neurosurgery, Ahead of Print: 1-23, 2024.

Objective: The National Football League (NFL) has seen increasing scrutiny regarding its handling of concussions, especially following an on-field incident involving Miami Dolphins' (™) Quarterback Tua Tagovailoa in the 2022 Season. The authors hoped to elucidate recent trends in the diagnosis and management of concussions over the course of five NFL seasons across 2019-2023.

Methods: The authors queried NFL injury reports from the 2019 through 2023 database recording players listed with Concussions. Weeks missed were calculated using NFL game logs. Player's concussions that did not occur in games, complicated by other injuries, or roster status were excluded.

Results: Searches of NFL injury reports resulted in the identification of 664 of 692 (96%) concussions that occurred in regular season games across the 2019-2023 seasons. Over the course of these five seasons 31% of players returned without missing a game, 39% of players missed one game, and 30% of

players missed two or games. No significant difference in the number of concussions per game or weeks missed was observed across the seasons observed. Players with concussions on teams that made the playoffs saw less weeks missed than those on non-playoff teams (0.86 v. 1.37, p=0.002).

Conclusions: Since the start of the 2021 NFL season, an increasing incidence of concussions has been noted, yet there was no change observed in the number of weeks missed following concussions. Trends in the rates of concussions across seasons remain largely stable, despite increased scrutiny over concussions in the sport.

Biceps femoris muscle-tendon strain during an entire overground sprint acceleration: A biomechanical explanation for hamstring injuries in the acceleration phase. Sports Biomechanics, Ahead of Print: 1-15, 2024.

Abstract - The objectives of this study were to analyze the peak muscle-tendon (MT) strain of the hamstring during an entire acceleration sprint overground and examine their relationship with relative joint angles and segment orientation in the sagittal plane, which are the direct causes of MT strain. Kinematic data were recorded using a 3D inertial motion capture system in 21 male semi-professional soccer players during 40-metre overground sprint. Scaled musculoskeletal models were used to estimate peak MT strain in the hamstring over 16 steps. Biceps femoris long head (BFLH) exhibited the largest peaks in MT strain compared to semitendinosus (ST) and semimembranosus (SM) muscles across all the steps, with its overall strain decreased as the number of steps and maximum speed increased. Hip flexion angle was found to be a strong predictor ($p < 0.001$) of joint angles, being the orientation of the pelvis in the sagittal plane of the segment with the greatest influence ($p < 0.001$) on the peak MT strain of BFLH during sprinting. The current study provides a biomechanical explanation for the high proportion of hamstring injuries in the acceleration phase of sprinting.

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Concurrent validity and test-retest reliability of VALD ForceDecks strength, balance, and movement assessment tests. Journal of Science and Medicine in Sport, Ahead of Print: 1-30, 2024.

Objectives: To evaluate the concurrent validity and test–retest reliability of common movement, strength, and balance tests using portable uniaxial dual force plates.

Design: Repeated measures cross-sectional study.

Methods: Sixteen healthy individuals participated in two testing sessions, where they performed 12 different movement, strength, and balance tests. Vertical ground reaction force and centre of pressure data were collected using the VALD ForceDecks simultaneously with ground-embedded laboratory force plates. Concurrent validity was assessed using root mean square error for raw time-series data and Bland–Altman plots for discrete metrics. Test–retest reliability was assessed using intraclass correlation coefficients and minimal detectable changes.

Results: ForceDecks recorded vertical ground reaction forces and center of pressure with high accuracy compared to laboratory force plates. The mean bias between systems was negligible (<2 N or 0.1 mm), with small limits of agreement (<5 N or 1 mm). Overall, 530/674 (79%) showed good or excellent validity (<10% difference) and 611/773 (79%) had good or excellent reliability (intraclass correlation coefficient >0.75). ForceDecks reliability was similar to laboratory force plates (<0.07 intraclass correlation coefficient median difference for all metrics).

Conclusions: Portable uniaxial force plates record highly accurate vertical ground reaction forces and center of pressure during a range of movement, strength, and balance tests. The VALD ForcDecks are a valid and reliable alternative to laboratory force plates when strict standardized testing and data analysis procedures are followed. Users should be aware of the validity and reliability characteristics of the tests and metrics they choose.

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Does a broad-spectrum cannabidiol supplement improve performance in a 10-min cycle ergometer performance test? European Journal of Sport Science, Ahead of Print: 1-8, 2024.

Abstract - Cannabidiol (CBD) is a non-intoxicating phytocannabinoid which has been proposed to possess anti-inflammatory and analgesic properties. Given the potential for perceptions of pain to limit exercise performance, the aim of the present study was to investigate if 3 weeks of daily CBD supplementation (150 mg day−1) improved performance in a 10-min performance-trial on a cycle ergometer. In a randomized, double-blind and placebo-controlled study, 22 healthy participants (n = 11 male and n = 11 female) completed two 10-min performance trials on a WattBike cycle ergometer interspersed with a 3-week supplementation period. Supplementation involved either 150 mg day−1 oral CBD or 150 mg day−1 of a visually identical placebo (PLA). During trials, ratings of perceived exertion (RPE [6–20]), heart rate (HR) and blood lactate (BLa) were collected every 2 min. Mean power (W) was also taken throughout the exercise at each time point. All data were analyzed using two- way ANOVAs. There were no significant differences (P > 0.05) between CBD or PLA groups for mean power (W) during the 10-min performance trial. There were also no significant differences (P > 0.05) in any of the physiological or perceptual parameters (HR, BLa and RPE) between conditions. Three weeks supplementation of a broad-spectrum CBD supplement did not improve performance via any change in RPE during a 10-min time trial on a cycle ergometer, and as such, this evidence does not support the claim that broad-spectrum CBD supplements could be performance-enhancing in this exercise modality.

Highlights

- 3 weeks supplementation of 150 mg/day broad-spectrum cannabidiol (CBD) does not in fluence perceived exertion during a 10-min aerobic performance trial.
- 3 weeks supplementation of 150 mg/day broad CBD does not improve physical performance in a 10-min aerobic performance trial.
- Athletes should continue abstinence of “Off-the-shelf” CBD supplements to avoid anti- doping rule violations from The World Anti-Doping Agency.

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Does it matter? Isometric or isokinetic assessment of quadriceps strength symmetry 9 months after ACLR in collegiate athletes. Sports Health, Ahead of Print: 1-9, 2024.

Background: Greater quadriceps strength symmetry is associated with better outcomes after anterior cruciate ligament reconstruction (ACLR). Isometric and isokinetic assessments of quadriceps strength inform therapeutic exercise prescription and return-to-sport decisions. It is unclear whether isometric and isokinetic measures provide similar information post-ACLR.

Hypothesis: Quadriceps strength symmetry is similar between isometric and isokinetic assessments. Isokinetic and isometric strength symmetries have similar associations to functional knee kinetics and self-reported knee function.

Study design: Cross-sectional study.

Level of evidence: Level 3.

Methods: NCAA Division I athletes (N = 35), 8.9 ± 2.5 months post-ACLR completed isometric and isokinetic quadriceps strength assessments, countermovement jumps (CMJs), and treadmill running. Self-reported knee function was assessed using the International Knee Documentation Committee Subjective Knee Form (IKDC). Agreement between isometric and isokinetic strength symmetry was assessed using Bland-Altman analysis, with associations to functional knee kinetics and IKDC assessed using Pearson correlations and linear regressions.

Results: Mean difference in quadriceps strength symmetry between isokinetic and isometric assessments was 1.0% (95% limits of agreement of -25.1% to 23.0%). Functional knee kinetics during running and CMJ were moderately to strongly associated with isometric strength symmetry ($r = 0.64$-0.80, $P < 0.01$) and moderately associated with isokinetic strength symmetry ($r = 0.41$-0.58, $P < 0.01$). IKDC scores were weakly to moderately associated with isometric ($r = 0.39$, $P = 0.02$) and isokinetic ($r = 0.49$, $P < 0.01$) strength symmetry.

Conclusion: Isokinetic and isometric assessments of quadriceps strength symmetry in collegiate athletes 9 months post-ACLR demonstrated strong agreement. Quadriceps strength symmetry is associated with functional knee kinetic symmetry post-ACLR.

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Clinical relevance: Considerable individual variation suggests mode of contraction should be consistent throughout postoperative assessment. Isometric strength symmetry may be a better indicator of functional knee kinetic symmetry, while isokinetic strength symmetry may be associated more closely with patient-reported outcomes.

Doping prevalence among U.S. elite athletes subject to drug testing under the World Anti-Doping Code. Sports Medicine – Open, 10:57, 2024.

Background: Determining the prevalence of doping within an elite athlete population is challenging due to the extreme sensitivity of the topic; however, understanding true doping prevalence is important when designing anti-doping programs and measuring their effectiveness. The objective of this study was to estimate the prevalence of doping among Olympic, Paralympic, World, and National-level competitive athletes in the United States subject to the World Anti-Doping Code. All athletes who were subject to the U.S. Anti-Doping Agency's Protocol for Olympic and Paralympic Movement Testing, a World Anti-Doping Code ("Code")-compliant anti-doping program, were invited to complete a web-delivered survey. Using a direct questioning approach, the survey items asked athletes whether they had used each specific category of banned substance / method on the World Anti-Doping Agency's Prohibited List. Multiple strategies to encourage honest reporting (e.g., protecting anonymity by collecting minimal demographic information; using an outside organization to administer the survey) and to detect inconsistent responses were used.

Results: Depending on the method of calculation, 6.5–9.2% of the 1,398 respondents reported using one or more prohibited substances or methods in the 12 months prior to survey administration. Specific doping prevalence rates for each individual substance / method categories ranged from 0.1% (for both diuretics / masking agents and stem cell / gene editing) to 4.2% for in-competition use of cannabinoids.

Conclusion: Determining the prevalence of doping within different athlete populations is critical so that sport governing bodies can evaluate their anti-doping efforts and better tailor their programming. By measuring doping prevalence of specific categories of substances and methods, rather than just the overall prevalence of doping, this study also highlights where sport governing bodies should focus their future educational and detection efforts.

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Key Points

• Estimated doping prevalence among U.S. elite athletes ranged from 6.5 to 9.2%.

• The most prevalent category of doping was in-competition use of cannabinoids (at 4.2%), whereas the least prevalent categories of doping were diuretics or masking agents (0.1%), stem cell or gene editing (0.1%), narcotics (0.2%), and hormone and metabolic modulators (0.2%).

• Of the athletes who reported doping, most reported using only one substance or method.

Effect of personalized sodium replacement on fluid and sodium balance and thermophysiological strain during and after ultraendurance running in the heat. International Journal of Sports Physiology and Performance, 19(2): 105-115, 2023.

Purpose: To investigate the effect of personalized sweat sodium replacement on drinking behavior, sodium and water balance, and thermophysiological responses during and after ultraendurance running in hot conditions.

Methods: Nine participants (7 male, 2 female) completed two 5-hour treadmill runs (60% maximum oxygen uptake, 30°C ambient temperature), in a double-blind randomized crossover design, consuming sodium chloride (SODIUM) capsules to replace 100% of previously assessed losses or placebo (PLACEBO). Fluid was consumed ad libitum.

Results: No effect of SODIUM was observed for ad libitum fluid intake or net fluid balance ($P > .05$). Plasma sodium concentration increased in both trials, but to a greater extent in SODIUM at 2.5 hours (mean [SD]: 4 [4] mmol·L-1 vs 1 [5] mmol·L-1; $P < .05$) and postexercise (4 [3] mmol·L-1 vs 1 [5] mmol·L-1; $P < .05$). Plasma volume change was not different between trials ($P > .05$) but was strongly correlated with sodium balance in SODIUM ($r = .880$, $P < .01$). No effect of sodium replacement was observed for heart rate, rectal temperature, thermal comfort, perceived exertion, or physiological strain index. During the 24 hours postexercise, ad libitum fluid intake was greater following SODIUM (2541 [711] mL vs 1998 [727] mL; $P = .04$), as was urinary sodium excretion (NaCl: 66 [35] mmol, Pl: 21 [12] mmol; $P < .01$).

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Conclusions: Personalized sweat sodium replacement during ultraendurance running in hot conditions, with ad libitum fluid intake, exacerbated the rise in plasma sodium concentration compared to no sodium replacement but did not substantially influence overall body-water balance or thermophysiological strain. A large sodium deficit incurred during exercise leads to substantial renal sodium conservation postexercise.

Effectiveness of recovery strategies after training and competition in endurance athletes: An umbrella review. Sports Medicine – Open, 10:55, 2024.

Background: Recovery strategies are used to enhance performance and reduce injury risk in athletes. In previous systematic reviews, individual recovery strategies were investigated to clarify their effectiveness for mixed groups of athletes. However, the current evidence is ambiguous, and a clear overview of (training) recovery for endurance athletes is still lacking.

Methods: We conducted an umbrella review based on a literature search in PubMed, Cochrane Database of Systematic Reviews, and Web of Science. Reviews published in English and before December 2022 were included. Systematic reviews and meta-analyses were eligible if they investigated the effectiveness of one or more recovery strategies compared with a placebo or control group after a training session in endurance athletes.

Results: Twenty-two reviews (nine systematic reviews, three meta-analyses, and ten systematic reviews with meta-analyses included) met the inclusion criteria. In total, sixty-three studies with 1100 endurance athletes were included in our umbrella review. Out of the sixty-three studies, eight provided information on training recovery time frame for data synthesis. Among them, cryotherapy and compression garments showed positive effects, while applying massage showed no effect. In general, none of the included recovery strategies showed consistent benefits for endurance athletes.

Conclusion: There is no particular recovery strategy that can be advised to enhance recovery between training sessions or competitions in endurance athletes. However, individual studies suggest that compression garments and cryotherapy are effective training recovery strategies. Further research should improve methodology and focus on the different time courses of the recovery process.

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Effects of chronic static stretching interventions on jumping and sprinting performance: A systematic review with multilevel meta-analysis. Frontiers in Physiology, 15:1372689, 2024.

Abstract - When improving athletic performance in sports with high-speed strength demands such as soccer, basketball, or track and field, the most common training method might be resistance training and plyometrics. Since a link between strength capacity and speed strength exists and recently published literature suggested chronic stretching routines may enhance maximum strength and hypertrophy, this review was performed to explore potential benefits on athletic performance. Based on current literature, a beneficial effect of static stretching on jumping and sprinting performance was hypothesized. A systematic literature search was conducted using PubMed, Web of Science and Google scholar. In general, 14 studies revealed 29 effect sizes (ES) (20 for jumping, nine for sprinting). Subgroup analyses for jump performance were conducted for short- long- and no stretch shortening cycle trials. Qualitative evaluation was supplemented by performing a multilevel meta-analysis via R (Package: metafor). Significant positive results were documented in six out of 20 jump tests and in six out of nine sprint tests, while two studies reported negative adaptations. Quantitative data analyses indicated a positive but trivial magnitude of change on jumping performance (ES:0.16, $p = 0.04$), while all subgroup analyses did not support a positive effect ($p = 0.09–0.44$). No significant influence of static stretching on sprint performance was obtained ($p = 0.08$). Stretching does not seem to induce a sufficient stimulus to meaningfully enhance jumping and sprinting performance, which could possibly be attributed to small weekly training volumes or lack of intensity.

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Epidemiology of injury and illness in North American professional men's soccer: Comparing COVID-19 lockdown with previous season. Sports Health, Ahead of Print: 1-7, 2024.

Background: After COVID-19 lockdown, studies across Europe and Asia examined its effect on professional soccer injury rates and severity; however, COVID-19 lockdown influence on injuries in United States men's professional soccer has not been evaluated.

Hypothesis: Injury and illness rates during the 2020 season were higher than the previous 2 seasons.

Study design: Retrospective observational cohort study.

Level of evidence: Level 3.

Methods: Medical staff from participating Major League Soccer (MLS) clubs entered injuries and athletic exposures during the study period into a league-wide electronic medical record system. Injury rate and severity were analyzed to examine differentials between the 2020 COVID season and historical controls.

Results: The injury incidence rate per 1000 hours in 2020 was 10.8, ie, higher than 2018 (5.4) and 2019 (5.0) ($P < 0.05$). Training injury incidence rate increased in 2020 (8.9) compared with 2018 (2.5) and 2019 (2.6) ($P < 0.05$), whereas match injury incidence decreased in 2020 (18.3) compared with 2018 (24.0) and 2019 (22.7) ($P < 0.05$). Incidence rates of lower extremity muscle injuries (6.04), anterior cruciate ligament (ACL) injuries (0.17), and concussions (0.49) were also higher in 2020 compared with 2018 (2.5, 0.07, 0.27) and 2019 (2.36, 0.05, 0.22) ($P < 0.05$). More injured players in 2020 missed >90 days (17.7%) than in 2018 (10.2%) and 2019 (10.1%) ($P < 0.05$). Incidence of all non-COVID-19 illness was higher in 2020 (3.93) than 2018 (1.53) and 2019 (1.32) ($P < 0.05$).

Conclusions: During 2020, there were significant increases in incidence rates of overall injuries, training injuries, lower extremity muscular injuries, ACL injuries, concussions, and non-COVID illness, along with a higher percentage of players missing >90 days compared with the 2 previous seasons.

Clinical relevance: These results may help clarify the effects of future MLS inseason work stoppages and periods of restricted training.

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Evaluating a 14-week neck strengthening protocol for neuromuscular indicators associated with head and neck trauma. Sports Medicine and Health Science, Ahead of Print: 1-7, 2024.

Abstract - Increased neck strength has been linked to a potential decrease in traumatic brain injuries (TBI). The purpose was to determine the efficacy of a neck-strengthening protocol using a novel neck-strengthening device to increase isometric neck strength and rate of force development (RFD). Utilizing self-generated centripetal force, participants trained for 14 weeks. A linear mixed model was used to analyze the relationship between post-assessment measurements and pre-assessments measurements, while accounting for repeated measure random effect at the individual level, and a regular random error term. RFD values were 4.344 times higher in the clockwise direction and 5.978 times higher in the counterclockwise direction when comparing pre and post assessment measurements. Isometric neck strength increased significantly ($p < 0.05$) in the cervical extension ($p = 0.010$) and left lateral flexion ($p = 0.009$) directions. The results can be used in strength training and clinical settings to potentially reduce the incidence of TBI.

Exercise cardiac load and autonomic nervous system recovery during in-season training: The impact on speed deterioration in American football athletes. Journal of Functional Morphology and Kinesiology, 8: 134, 2023.

Abstract - Fully restoring autonomic nervous system (ANS) function is paramount for peak sports performance. Training programs failing to provide sufficient recovery, especially during the in-season, may negatively affect performance. This study aimed to evaluate the influence of the physiological workload of collegiate football training on ANS recovery and function during the in-season. Football athletes recruited from a D1 college in the southeastern US were prospectively followed during their 13-week “in-season”. Athletes wore armband monitors equipped with ECG and inertial movement capabilities that measured exercise cardiac load (ECL; total heartbeats) and maximum running speed during and baseline heart rate (HR), HR variability (HRV) 24 h post-training. These metrics represented physiological load (ECL = HR·Duration), ANS function, and recovery, respectively. Linear regression models evaluated the associations between ECL, baseline HR, HRV, and maximum running speed.

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Athletes (n = 30) were 20.2 ± 1.5 years, mostly non-Hispanic Black (80.0%). Negative associations were observed between acute and cumulative exposures of ECLs and running speed ($\beta = -0.11 \pm 0.00$, $p < 0.0000$ and $\beta = -0.15 \pm 0.04$, $p < 0.0000$, respectively). Similarly, negative associations were found between baseline HR and running speed ($\beta = -0.45 \pm 0.12$, 95% CI: −0.70, −0.19; $p = 0.001$). HRV metrics were positively associated with running speed: (SDNN: $\beta = 0.32 \pm 0.09$, $p < 0.03$ and rMSSD: $\beta = 0.35 \pm 0.11$, $p < 0.02$). Our study demonstrated that exposure to high ECLs, both acutely and cumulatively, may negatively influence maximum running speed, which may manifest in a deteriorating ANS. Further research should continue identifying optimal training: recovery ratios during off-, pre-, and in-season phases.

Exercising more than 150 min/wk after concussion is associated with sleep quality improvements. Journal of Head Trauma Rehabilitation, Ahead of Print: 1-9, 2023.

Objective: To examine whether a high volume of aerobic exercise after concussion (>150 min/wk) is associated with improved sleep quality over a 1-month period. We hypothesized that more than 150 min/wk of exercise would be associated with improved sleep quality across concussion recovery. Design: Prospective cohort observational study.

Setting: Sports medicine clinic.

Participants: Adolescents initially tested 8.4 ± 3.5 (range, 2-18) days postconcussion who returned for a follow-up assessment 34.3 ± 7.7 (range: 20-49) days postconcussion.

Main Outcome Measures: Participants completed the Pittsburgh Sleep Quality Index and the Post-Concussion Symptom Inventory. No specific exercise or sleep recommendations were given beyond what their treating physician provided. Between study visits, participants recorded exercise performed via wrist-worn actigraphy. We calculated average exercise minutes per week and grouped participants as those who exercised more than 150 min/wk versus those who exercised 150 min/wk or less.

Results: Thirty-six adolescents participated. Fifteen (42%) recorded more than 150 min/wk of aerobic exercise (age = 14.0 ± 1.7 years; 47% female; mean = 5.6 ± 1.2 d/wk of exercise; mean = 49.2 ± 17.5

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min/session), and 21 recorded 150 min/wk or less of aerobic exercise (age = 15.0 ± 1.9 years; 76% female; mean = 2.7 ± 1.6 d/wk of exercise; mean = 30.2 ± 7.8 min/session). There were no significant group differences in the proportion of those who self-reported beginning physical activity prior to enrollment (47% vs 33%; P = .42) or for initial sleep quality rating (8.0 ± 3.7 vs 8.6 ± 4.1; P = .67) or initial concussion symptom severity rating (34.9 ± 28.0 vs 42.6 ± 25.9; P = .40). The group that exercised more than 150 min/wk between visits demonstrated significantly greater median PSQI rating improvements than those who exercised 150 min/wk or less, with a large effect size noted (median change [interquartile range] = 5 [3, 7] vs 1 [0, 4]; P = .008; Cohen d = 0.96).

Conclusion: Current recommendations suggest that subsymptom aerobic exercise can be beneficial after concussion. Our findings indicate that an exercise volume of more than 150 min/wk led to greater sleep quality improvements than those who exercised below this level.

From high-speed running to hobbling on crutches: A machine learning perspective on the relationships between training doses and match injury trends. Sports Performance & Science Reports, Vol. 1: 1-11, 2024.

Aim: The present study aimed to bring a fresher and deeper look at the association between HSR volumes during the training week and non-contact muscle injuries during the match ending the microcycle. Embracing a broader perspective, our present study, building upon earlier research on max speed exposures (8) and rest days (9), amalgamates data from a wide array of clubs. This expansive approach not only enhances our sample size but also enriches our analysis with diverse contexts. Distinguishing itself from previous studies, we, for the first time, integrate machine learning methods to unearth deeper pat terns in football training and injury correlation. While past studies like those by L´opez-Valenciano (19), Ayala (1), Haller (18), and Rossi (25) have employed machine learning models for injury forecasting, their relevance was limited due to two primary factors. Firstly, these studies did not boast the extensive and varied dataset we utilized. L´opez-Valenciano (19) combined soccer and handball players, a mix that could lead to misleading conclusions due to the different training regimens and physiological characteristics of these athletes. Ayala (1) focused on just four professional soccer teams, Haller (18) on a single youth soccer team, and Rossi (25) on one professional team, thereby

restricting the breadth of their findings. Secondly, previous analyses often yielded predictive insights without tangible action plans, lacking in actionable outcomes for practitioners. Furthermore, they did not delve into the dynamics of match turnaround, a critical aspect of how coaches periodize training (7,8,9). Our study provides for the first-time actionable insights into the complex dynamics of training periodization and injury risk. By doing so, we aim to equip practitioners with comprehensive tools to fine-tune training programs, ultimately aiming to diminish match injuries and enhance player safety and performance. Our inclusive and technologically sophisticated approach offers a more nuanced understanding of the variegated football environments, capturing the intricacies of this sport in a unified and insightful narrative.

Height restrictions for dual-energy x-ray absorptiometry: What are our options for body composition and bone health precision? Journal of Strength and Conditioning Research, Ahead of Print: 1-7, 2024.

Abstract - Dual-energy X-ray absorptiometry (DXA) is commonly used for testing athlete's body composition, but many athletes are too tall. The first aim was to assess the viability of combining upper- and lower-body regions of interest (ROI), creating a combined scan for tall athletes who do not fit on the table and second, to provide practical solutions to DXA users who routinely scan tall athletes. Sixty subjects (34.8 ± 11.9 years; 171 ± 9 cm) completed 2 total-body DXA scans for baseline precision testing, using GE Lunar Prodigy (LP) or Hologic Horizon A (HA) models. Next, an upper body ROI from the skull to the distal femoral condyles was combined with a flipped scan (feet-to-head) ROI encompassing the proximal tibial plateau to the distal foot. Soft and bone tissue coefficient of variance (CV%) were calculated between the baseline scans and for the newly combined ROI scan. The combined ROI scan added 0.25-0.63% and 0.01-4.35% error rates for the LP and HA, respectively. An exploratory assessment of a GE Lunar iDXA demonstrated results similar to the HA with 4%+ error. The combined ROI scan is a user-friendly and precise method for older LP systems adding less than 1% error; however, newer DXA systems cannot use the stitched scan technique. Coaches and practitioners who use newer DXAs must prioritize consistently using the same boney landmarks (head, jaw, or feet) and ROI heights to provide precise longitudinal assessments of tall athletes' bone and body composition, until larger DXA tables become available.

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Practical Applications - The use of DXA systems in athletics is only gaining popularity; however, athletes who are taller than 193 cm often have tissue omitted across multiple time points, reducing the data's utility. There are 3 main considerations: First, this is one of the few studies to evaluate a practitioner-friendly scanning technique that guides users through ROI creation to allow for precise data collection in tall athletes over time. Second, coaches, staff, and practitioners with the HA or iDXA must carefully consider the limitations discussed here when creating their team's standard scanning procedures for tall athletes. Finally, DXA users need to be able to discuss their athlete's data over time within the context of their specific machine's precision. If the DXA being used has not had baseline precision testing, then serial assessments are suspect to inaccurate and potentially dangerous interpretations.

Impact of melatonin supplementation on sports performance and circulating biomarkers in highly trained athletes: A systematic review of randomized controlled trials. Nutrients, 16:1011, 2024.

Abstract - Melatonin (N-acetyl-5 methoxytryptamine) is an indolic neurohormone that modulates a variety of physiological functions due to its antioxidant, anti-inflammatory, and immunoregulatory properties. Therefore, the purpose of this study was to critically review the effects of melatonin supplementation in sports performance and circulating biomarkers related to the health status of highly trained athletes. Data were obtained by performing searches in the following three bibliography databases: Web of Science, PubMed, and Scopus. The terms used were "Highly Trained Athletes", "Melatonin", and "Sports Performance", "Health Biomarkers" using "Humans" as a filter. The search update was carried out in February 2024 from original articles published with a controlled trial design. The PRISMA rules, the modified McMaster critical review form for quantitative studies, the PEDro scale, and the Cochrane risk of bias were applied. According to the inclusion and exclusion criteria, 21 articles were selected out of 294 references. The dose of melatonin supplemented in the trials ranged between 5 mg to 100 mg administered before or after exercise. The outcomes showed improvements in antioxidant status and inflammatory response and reversed liver damage and muscle damage. Moderate effects on modulating glycemia, total cholesterol, triglycerides, and creatinine were reported. Promising data were found regarding the potential benefits of melatonin in hematological biomarkers, hormonal responses, and sports performance. Therefore, the true efficiency of melatonin to directly

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improve sports performance remains to be assessed. Nevertheless, an indirect effect of melatonin supplementation in sports performance could be evaluated through improvements in health biomarkers.

Injury in starting and replacement players from five professional men's Rugby Unions. Sports Medicine, Ahead of Print: 1-11, 2024.

Objectives: The aim of this study was to compare the incidence, severity, and burden of injury in starting and replacement players from professional men's teams of five rugby unions.

Methods: Match injuries of greater than 24 h time-loss (including data on the severity, match quarter, event, body region) and player minutes of match exposure data were collated for all starting and replacement players in the men's English Premiership, Welsh Pro14 (both 2016/17-2018/19 seasons), and Australian, New Zealand, and South African Super Rugby (all 2016-2018 seasons) teams. Injury incidences and mean injury burden (incidence × days missed) were calculated, and rate ratios (RRs) (95% confidence intervals [CIs]) were used to compare injury incidence and burden between starting (reference group) and replacement players.

Results: Overall injury incidence was not different between starters and replacements for all injuries (RR = 0.98, 95% CI 0.88-1.10), nor for concussions (RR = 0.85; 95% CI 0.66-1.11). Mean injury burden was higher for replacement players (RR = 1.31, 95% CI 1.17-1.46). Replacement injury incidence was lower than the starters in the third (RR = 0.68, 95% CI 0.51-0.92) and fourth (RR = 0.78, 95% CI 0.67-0.92) match quarters. Injury incidence was not different between starters and replacements for any match event or body region, but compared with starters, replacements' injury burden was higher in lower limbs (RR = 1.24, 95% CI 1.05-1.46) and in the tackled player (RR = 1.30, 95% CI 1.01-1.66).

Conclusion: This study demonstrated a lower injury incidence in replacement players compared with starters in the second half of matches, with a higher injury burden for replacement players due to higher mean injury severity.

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Key Points

- The number of match minutes or total number of replacements used by one team did not impact on the number or burden of injuries sustained by the opposing team.
- Injury incidence rates over the course of an entire match were not different between replacement and starting players, but incidence rates were lower for replacements compared with starters in the second half of matches.
- This study provides objective data on which the governing body may make informed decisions with respect to starting players versus replacement players, but the current results do not support changes to the current laws on the use of replacements in the game.

Lack of proprioceptive strategy modulation leads to at-risk biomechanics for anterior cruciate ligament in healthy athletes. Medicine and Science in Sports and Exercise, Vol. 56(5): 942-952, 2024.

Introduction: Anterior cruciate ligament (ACL) injuries are frequent in handball, and altered sensory integration may contribute to increased injury risk. Recent evidence showed that proprioceptive postural control strategies differ among athletes. The aim of this study was to evaluate the relationship between proprioceptive strategy and biomechanics during side-cutting maneuvers.

Methods: A total of 47 handball players performed anticipated and unanticipated cutting tasks. Their postural proprioceptive strategy was then characterized according to the perturbation of the center of pressure displacement generated by the muscle vibration on a firm and foam surface. Individuals able to reweight proprioception from ankle to lumbar signals according to the stability of the support were defined as flexible. Conversely, athletes maintaining an ankle-steered strategy on foam surface were characterized as rigid. Statistical parametric mapping analysis was used to compare pelvic and lower limb side-cutting kinematics, kinetics, and EMG activity from seven muscles 200 ms before and after initial contact (IC) using a two-way ANOVA (group-condition).

Results: Twenty athletes (11 females and 9 males, 18.5 yr) were characterized as flexible and 20 athletes (12 females and 8 males, 18.9 yr) as rigid. No interaction between condition and proprioceptive profile

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was observed. More ipsilateral pelvic tilt before IC and lower vastus lateralis (VL) activity immediately after IC was observed during CUT ant . When comparing proprioceptive strategy, rigid individuals exhibited less preactivity of the semitendinosus ($P < 0.001$) and higher VL activity ($P = 0.032$). Conversely, rigid showed higher gluteus medius preactivity ($P < 0.05$) and higher VL activity 100 ms after IC ($P < 0.001$). Ankle was also more internally rotated before and during the stance phase ($P < 0.05$) among rigid athletes.

Conclusions: Rigid handball players exhibited at-risk determinants for anterior cruciate ligament injuries during side-cutting maneuvers.

Limited return to preinjury performance in NCAA Division 1 American football players with hamstring injuries. Orthopaedic Journal of Sports Medicine, 12(5): 1-6, 2024.

Background: Hamstring strains are common among elite athletes, but their effect on return to the same level of play in American football has been incompletely characterized.

Purpose: Data on National Collegiate Athletics Association Division I college football players with acute hamstring strains were gathered to identify the effects these injuries have on both return to play and athletic performance regarding velocity, workload, and acceleration.

Study design: Case Series; Level of evidence, 4.

Methods: Injury data for a single Division I football team were prospectively recorded over a 4-year period. Players wore global navigation satellite system and local positioning system (GNSS/LPS) devices to record movement data in practices and games. The practice and game data were cross-referenced to evaluate players with isolated acute hamstring strains. Comparisons were made regarding players' pre- and postinjury ability to maintain high velocity (>12 mph [19.3 kph]), maximal velocity, triaxial acceleration, and inertial movement analysis (IMA). There were 58 hamstring injuries in 44 players, of which 25 injuries from 20 players had GNSS/LPS data.

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Results: Players were able to return to play from all 25 injury incidences at a mean of 9.2 days. At the final mean follow-up of 425 days, only 4 players had reached preinjury function in all measurements; 12 players were able to return in 2 of the 4 metrics; and only 8 players reached their preinjury ability to maintain high velocity. For those who did not achieve this metric, there was a significant difference between pre- and postinjury values (722 vs 442 m; P = .016). A total of 14 players were able to regain their IMA. Players who returned to prior velocity or acceleration metrics did so at a mean of 163 days across all metrics.

Conclusion: While players may be able to return to play after hamstring strain, many players do not reach preinjury levels of acceleration or velocity, even after 13.5 months. Further studies are needed to confirm these findings, assess clinical relevance on imaging performance, and improve hamstring injury prevention and rehabilitation.

Longitudinal changes in quadriceps morphology over the first 3 months after anterior cruciate ligament reconstruction. Medicine and Science in Sports and Exercise, Vol. 56(4): 933-941, 2024.

Purpose: Neuromuscular deficits and atrophy after anterior cruciate ligament reconstruction (ACLR) may be accompanied by changes in muscle composition and poor quadriceps muscle quality (QMQ). Quadriceps atrophy occurs after ACLR but improves within the first three postoperative months, yet this hypertrophy could be attributable to increases in noncontractile tissue (i.e., poor QMQ). The purposes of this study were to evaluate changes in QMQ after ACLR and to determine if changes in QMQ and cross-sectional area (CSA) occur in parallel or independently.

Methods: A longitudinal prospective cohort design was implemented to evaluate QMQ and CSA in 20 individuals with ACLR and 12 healthy controls. Participants completed three testing sessions (baseline/presurgery, 1 month, and 3 months) during which ultrasound images were obtained from the vastus lateralis (VL) and rectus femoris (RF). QMQ was calculated as the echo intensity (EI) of each image, with high EI representing poorer QMQ. Anatomical CSA was also obtained from each image.

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Results: RF and VL EI were greater at 1 and 3 months in the ACLR limb compared with baseline and the contralateral limb and did not change between 1 and 3 months. VL and RF CSA in the ACLR limb were smaller at 1 and 3 months compared with the contralateral limb and controls (VL only) but increased from 1 to 3 months. Changes in QMQ and CSA were not correlated.

Conclusions: QMQ declines within the first month after ACLR and does not improve by 3 months although hypertrophy occurs, suggesting that these morphological characteristics change independently after ACLR. Poorer QMQ represents greater concentration of noncontractile tissues within the muscle and potentially contributes to chronic quadriceps dysfunction observed after ACLR.

Lower limb joint mechanics during maximal accelerative and decelerative running. Medicine and Science in Sports and Exercise, Ahead of Print: 1-55, 2024.

Introduction: Maximal acceleration and deceleration tasks are frequently required in team sports, often occurring rapidly in response to external stimuli. Accelerating and decelerating can be associated with lower limb injuries, thus knowledge of joint mechanics during these tasks can improve the understanding of both human high performance and injury mechanisms. The current study investigated the fundamental differences in lower limb joint mechanics when accelerating and decelerating by directly comparing the hip, knee and ankle joint moments and work done between the two tasks.

Methods: Twenty participants performed maximal effort acceleration and deceleration trials, with three-dimensional marker trajectories and ground reaction forces collected simultaneously. Experimental data was combined with inverse dynamics analysis to compute joint moments and work.

Results: Net joint work for all lower limb joints was positive during acceleration, and negative during deceleration. This occurred due to significantly greater positive work production from the ankle and hip during acceleration, and significantly greater negative work production from all joints during deceleration. The largest contributions to positive work during acceleration came from the ankle, followed by the hip and knee joints; whilst the largest contributions to negative work during deceleration came from the knee and hip joints, followed by the ankle. Peak joint moments were

significantly greater when decelerating compared to accelerating, except for the peak ankle plantarflexion and hip flexion moments which were significantly greater when accelerating.

Conclusions: Our findings may help to guide training interventions which aim to enhance the performance of acceleration and deceleration tasks, whilst also mitigating the associated injury risk.

Editorial: Marketing of unproven and unapproved regenerative medicine therapies. Sports Health, 16(3): 312-314, 2024.

You may have received the same blast email that I have received a number of times from a company in Scottsdale, Arizona, that opens with "End of Year Blowout Sale: 3 for 1 on Exosomes or Whartons Jelly." The email further states that these products are "manufactured in an FDA registered, cGMP compliant, ISO Certified lab," despite the fact that such birth products can be used only in a registered FDA clinical trial, since a directive from the United States (US) Food and Drug Administration (FDA) on May 31, 2021. This is a good example of the aggressive, direct-to-consumer marketing of unapproved and unproven cell therapies. As I am writing this editorial, I see this headline from MedPage Today, January 22, 2024: "FDA Warns Exosome Maker for Marketing Outside Its IND". According to this article, the manufacturer received an Investigational New Drug (IND) for a phaseI/IIa clinical trial evaluating a single intravenous dose of a drug to treat COVID-19. However, the company apparently advertised their product for numerous other conditions outside of this specific IND. Furthermore, the FDA cited the company for "significant deviations" from current good manufacturing practice requirements as well as issues with sterility testing of their product. The 2 instances documented above point out some of the challenges in the rapidly growing regenerative medicine arena. The allure of regenerative medicine has led to a global industry of direct-to-consumer offerings of biologic products that, despite their promising potential, often have very little clinical data to support their use. This has been a pervasive problem in the area of regenerative medicine and is only growing. Regenerative medicine and orthobiologics can be lucrative, leading to aggressive marketing of products that have not been tested adequately for safety and/or efficacy. In this editorial I will explain some of the tactics used in these persuasive and aggressive marketing schemes. I will then briefly describe the criteria that should be considered before a product might be considered "proven", and finally I will list some mechanisms that can be used to report deceptive advertising claims and inappropriate marketing of unproven therapies.

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MMP3 single-nucleotide polymorphisms are associated with noncontact ACL injuries in competing high-level athletes. Journal of Orthopaedic Research, 42: 109-114, 2024.

Abstract - Matrix metalloproteinases (MMPs) play an important role in matrix remodeling, as well as in ligament integrity. Anterior cruciate ligament (ACL) rupture is a severe and frequent knee injury in sports. The aim of this study was to investigate polymorphisms within the MMP3 gene with the predisposition for noncontact ACL rupture in the Croatian professional athletes. One hundred eighty-seven (95 with ACL rupture occurring through a noncontact mechanism and 92 asymptomatic controls) unrelated Caucasians were recruited between 2016 and 2017. All participants were genotyped for three single-nucleotide polymorphisms (SNP) within the MMP3 gene: rs591058 C/T, rs650108 A/G, and rs679620 G/A using the pyrosequencing method. For all three investigated SNPs, genotype frequencies have significantly differed between cases and controls. The MMP3 rs591058 TT ($p = 0.0012$, odds ratio [OR] = 38.541, 95% confidence interval [CI] = 1.7024–8.7254), rs650108 GG ($p = 0.0051$, OR = 23.338, 95% CI = 1.2899–4.2226) and rs679620 AA ($p = 0.0030$, OR = 34.750, 95% CI = 1.5266–7.9101) genotypes, as well as haplotype variant T-G-A ($p = 0.0104$, OR = 1.71, 95% CI = 1.13–2.59) were significantly overrepresented in cases compared to controls. These results support association between functional variants within the MMP3 gene and the risk of ACL rupture. Still, further research is needed to corroborate these results in a larger population.

MRI findings prior to return to play as predictors of reinjury in professional athletes: A novel decision-making tool. Insights into Imaging, 13:203, 2022.

Background: Because MRI has shown great accuracy in assessing acute muscle injuries, identification of risk factors for reinjury before return to play (RTP) in professional athletes during the healing process could be very relevant. We assessed the value of MRI findings prior to RTP as predictors of reinjury.

Methods: Retrospective observational study of 59 professional athletes, mean age 26 years, with first-time acute muscle injury and successful rehabilitation ready to RTP. They underwent MRI within 6 days of the injury and within 7 days prior to RTP. The primary outcome was reinjury. Risk of reinjury was

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assessed using radiological signs in control MRI scans before RTP. The risk was classified as low, medium or high when none, one or two radiological signs were observed, respectively.

Results: Reinjury occurred in 9 participants, with a rate of 15.2%. None of the baseline MRI-related variables was significantly associated with reinjury. In the control MRI scan performed within 7 days prior to RTP, three independent findings were significantly associated with reinjury. These included transversal and/or mixed connective tissue gap ($p = 0.002$), intermuscular oedema ($p = 0.015$) and callus gap ($p = 0.046$). In the predictive model of the risk of reinjury, the presence of two of these radiological signs, together with interstitial feathery oedema, was associated with a high risk of recurrence (OR 29.58, 95% CI 3.86-226.64; $p = 0.001$).

Conclusions: In professional athletes with acute muscle injuries of the lower limbs successfully rehabilitated, some radiological signs on MRI performed shortly before RTP were associated with a high risk of reinjury.

Key points

• Certain radiological findings of acute muscle injury on MRI performed shortly prior to return to play may be considered risk factors of reinjury.

• Tere is a high risk of reinjury when two of these five radiological findings are present (connective tissue gap, loss of tendon tension, intermuscular oedema, callus gap and interstitial feather oedema).

• Despite appropriate progression during rehabilitation, MRI studies appear necessary in the decision to return to competition

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Muscle activity and kinematics during three hamstring strengthening exercises compared to sprinting: A cross-sectional study. International Journal of Sports Physical Therapy, 19(5): 569-580, 2024.

Background: During sprinting, the biceps femoris long head predominantly gets injured, while hamstring strengthening exercises predominantly activate the semitendinosus more effectively. Understanding how joint dominance influences hamstring activity may offer clarity on appropriate exercise selection in strengthening programs.

Purpose: This study compared three hip-dominant hamstring exercises: the rocker, perpetuum mobile fast and slow (PMfast and PMslow) and the Nordic Hamstring exercise (NHE) on their potential to simulate sprint-like activity and kinematics.

Methods: Muscle activity of the posterior kinetic chain (biceps femoris, semitendinosus, gluteus maximus, and medial gastrocnemius) was measured with surface electromyography (sEMG) during the exercises and treadmill running at 75% of the individual maximal sprint velocity in male athletes. sEMG data were normalized to maximal sprinting. 3D-motion capture was employed to assess hip and knee angles.

Results: Eight male athletes were included (age: 24.0 years ± SD 2.9; body mass: 76.8 kg ± 7.7; height: 1.79 m ± 0.08). Greater activity of the hamstrings occurred during the explosive exercises ranging from 63.9% [95%CI: 56.3-71.5%] (rocker) to 49.0% [95%CI: 40.4-57.6%] (PMfast) vs. 34.0% [95%CI: 29.1-38.9%] (NHE) to 32.1% [95%CI: 26.9-37.3%] (PMslow). The rocker showed greatest hamstring and gluteus maximus activity. Biceps femoris consistently showed greater activity than the semitendinosus across all exercises in peak (mean difference: 0.16, [95%CI: 0.07-0.26]) and average (mean difference: 0.06, [95%CI: 0.01-0.11]) activity. PMfast, PMslow and NHE demonstrated less hip flexion angle at peak hamstring activity than the rocker and high-speed running and every exercise showed less hamstring elongation stress than during high-speed running.

Discussion: Hamstring activity is comparable to high-intensity treadmill running for NHE and PMslow, and greater for the rocker and PMfast. Gluteus maximus activity varied, with the rocker and PMfast showing greater activity than in sprinting. All examined exercises demonstrated their peak activity at short hamstring muscle length.

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Musculotendinous ruptures of the Achilles tendon had greater heel-rise height index compared with mid-substance rupture with non-operative management: A retrospective cohort study. Journal of ISAKOS, 9(2): 148-152, 2024.

Introduction: Achilles tendon ruptures (ATRs) may occur at varying locations with ruptures at the mid-substance (MS) of the tendon most common, followed tears at the musculotendinous (MT) junction. There is scant literature about the outcome of MT ATR. This study compared the outcome of patients with a MT ATR with patients following a MS ATR.

Methods: The diagnostic features and clinical outcome of 37 patients with a MT ATR were compared with a cohort of 19 patients with a MS ATR. Patients in both groups were managed non-operatively and received the same rehabilitation protocol with weight-bearing rehabilitation in protective functional brace.

Results: From February 2009 to August 2023, 556 patients presented with an ATR. Of these, 37 (6.7 %) patients were diagnosed with a MT tear. At final follow-up, at 12 months following injury, the MT group reported an Achilles tendon total rupture score (ATRS) of mean (standard deviation (SD)) of 83.6 (3.5) (95 % confidence interval (CI) 81.8, 85.4) and median (inter-quartile range (IQR)) ATRS of 86 points (78-95.5) and the MS group mean (SD) of 80.3 (8.5) (95%CI) 76.1, 80.5) and median (IQR) of 87 points (59-95) ($p = 0.673$). Functional evaluation, however, revealed statistically significant differences in mean (SD) heel-rise height index MT group 79 % (25) (95%CI 65.9, 92.1) and MS group 59 % (13) (95%CI 51.9, 67.1) ($p = 0.019$). In the MT rupture group, there were considerably less complications than the MS rupture group.

Conclusions: When managed non-operatively, with only a 6 weeks period of brace protection, patients have little limitation although have some residual reduction of single heel-rise at the one-year following MT ATR.

Level of evidence: IV.

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Observation of risk of concussion following 'lowering the helmet to initiate contact' penalties in the NFL. Journal of Orthopaedic Reports, 100347, 2024.

Objective: This observational study examines the associated risk for concussion following 'lowering the helmet to initiate contact' penalties in the National Football League.

Methods: Using officially published injury data on the National Football League (NFL) website, the authors reviewed NFL Players diagnosed with a concussion from 2015 to 2022 and who received a 'lowering the helmet to initiate contact' penalty from 2018 to 2020. Reported concussions were evaluated for association with 'lowering the helmet to initiate contact' penalties. The Friedman test assessed overall differences between the years. Pairwise comparisons were performed using the Wilcoxon signed-rank test with a Bonferroni correction to adjust for multiple comparisons.

Results: The average incidence of all concussions per play was 0.45% for regular-season games. From 2018 to 2020, 86 lowering the helmet to initiate contact penalties were called. Three of 86 penalties resulted in concussions, with a 3.5% incidence. The Friedman test indicated significant differences in concussion trends across the years ($\chi 2 = 16.0$, $p = 0.0003$). Pairwise analysis with the Wilcoxon signed-rank test and adjusted for multiple comparisons with a Bonferroni correction ($\alpha = 0.05/28 = 0.0018$) showed no statistically significant differences between years. There was a significant negative correlation of overall concussion incidence per year.

Conclusions: The incidence of concussion with players penalized for lowering the helmet to initiate contact is greater than the overall incidence of concussions in all plays. While there is no difference in total concussions that occur, the concussion rate has a significant negative trend between 2015 and 2022. However, it is difficult to analyze the nonimmediate effects of this penalty on players' intended play and practice, which may further decrease concussions.

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One week of single leg immobilization lowers muscle connective protein synthesis rates healthy, young adults. Medicine and Science in Sports and Exercise, Ahead of Print: 1-40, 2023.

Purpose: Short periods of limb immobilization lower myofibrillar protein synthesis rates. Within skeletal muscle, the extracellular matrix of connective proteins is recognized as an important factor determining the capacity to transmit contractile force. Little is known regarding the impact of immobilization and subsequent recovery on muscle connective protein synthesis rates. This study examined the impact of one week of leg immobilization and two weeks of subsequent ambulant recovery on daily muscle connective protein synthesis rates.

Methods: Thirty healthy, young (24±5 y) men were subjected to 7 days of one-legged knee immobilization followed by 14 days of ambulant recovery. Deuterium oxide ingestion was applied over the entire period and muscle biopsy samples were collected before immobilization, after immobilization, and after recovery to measure muscle connective protein synthesis rates and mRNA expression of key extracellular matrix proteins (collagen I, collagen III), glycoproteins (fibronectin, tenascin-C), and proteoglycans (fibromodulin, and decorin). A two way repeated measures (time x leg) ANOVA was used to compare changes in muscle connective protein synthesis rates during immobilization and recovery.

Results: During immobilization, muscle connective protein synthesis rates were lower in the immobilized (1.07±0.30 %/d) compared with the non-immobilized (1.48±0.44 %/d; $P<0.01$) leg. When compared to the immobilization period, connective protein synthesis rates in the immobilized leg increased during subsequent recovery (1.48±0.64 %/d; $P<0.01$). Following recovery, skeletal muscle collagen I, collagen III, fibronectin, fibromodulin, and decorin mRNA expression increased when compared to the post-immobilization timepoint (all $P<0.001$).

Conclusions: One week of leg immobilization lowers muscle connective protein synthesis rates. Muscle connective protein synthesis rates increase during subsequent ambulant recovery, which is accompanied by increased mRNA expression of key extracellular matrix proteins.

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Oxygen straight to the brain: An overview of hyperbaric oxygen therapy for a variety of brain morbidities. Current Sports Medicine Reports, 23(4): 13-136, 2024.

Abstract - Hyperbaric oxygen therapy as a treatment for conditions like traumatic brain injury, posttraumatic stress disorder, and migraines would seem intuitive, given its effect on condition-related ischemia and inflammation. However, hyperbaric therapeutic impacts for these in acute and chronic, or prolonged symptoms are elusive. This narrative review of hyperbaric's utility provided in sections per disease renders first a review of conventional pathological mechanisms and then articulates hyperbaric treatment targets versus their respective impacts. Multiple challenges exist using hyperbaric oxygen therapy for each morbidity, even in tertiary and adjunctive treatments. An almost universal shortfall across studies includes a lack of consistent, appropriate patient selection criteria intersected with delivery timing of therapy to symptomatic target, necessary to provide a higher fidelity in treatment metrics. Further research into these respective conditions is needed along with a revisitation of hyperbaric oxygen therapy's application to their conventional pathological mechanisms, lending new perspective to their employment and efficacy.

Partially constrained group variable selection to adjust for complementary unit performance in American college football. Journal of Applied Statistics, Ahead of Print: 1-16, 2024.

Abstract - Given the importance of accurate team rankings in American college football (CFB) – due to heavy title and playoff implications – strides have been made to improve metrics for team performance evaluation, going from basic averages (e.g. points scored per game) to metrics that adjust for a team's strength of schedule, but one aspect that's yet to be accounted for is the ability of team's offense and defense to complement one another, termed 'complementary football'. American football is unique because the same team's offensive and defensive units typically consist of separate player sets that don't share the field simultaneously, which tempts one to evaluate them independently. Yet, some aspects of your team's defensive (offensive) performance may directly impact the complementary unit,

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e.g. turnovers forced by your defense could lead to easier scoring chances for your offense. Our main goal is to identify the most consistently influential features of complementary football in a data driven way, subsequently adjusting each team's offensive (defensive) performance for that of their complementary unit. To achieve that, for the 2009–2019 CFB seasons∗, we incorporate natural splines with group penalty approaches, conducting partially constrained optimization to guarantee the full adjustment for the strength of schedule and home-field factor.

Prevalence of cardiovascular disease and risk factors among former National Football League players. Journal of the American College of Cardiology, 83(19): 1827-1837, 2024.

Background: Cardiovascular disease (CVD) is the leading cause of death worldwide, but prevalence estimates in former professional athletes are limited.

Objectives: HUDDLE (Heart Health: Understanding and Diagnosing Disease by Leveraging Echocardiograms) aimed to raise awareness and estimate the prevalence of CVD and associated risk factors among members of the National Football League (NFL) Alumni Association and their families through education and screening events.

Methods: HUDDLE was a multicity, cross-sectional study of NFL alumni and family members aged 50 years and older. Subjects reported their health history and participated in CVD education and screening (blood pressure, electrocardiogram, and transthoracic echocardiogram [TTE] assessments). Phone follow-up by investigators occurred 30 days postscreening to review results and recommendations. This analysis focuses on former NFL athletes.

Results: Of 498 participants screened, 57.2% (N ¼ 285) were former NFL players, the majority of whom were African American (67.6%). The prevalence of hypertension among NFL alumni was estimated to be 89.8%, though only 37.5% reported a history of hypertension. Of 285 evaluable participants, 61.8% had structural cardiac abnormalities by TTE. Multivariable analysis showed that hypertension was a significant predictor of clinically relevant structural abnormalities on TTE.

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Conclusions: HUDDLE identified a large discrepancy between participant self-awareness and actual prevalence of CVD and risk factors, highlighting a significant opportunity for population health interventions. Structural cardiac abnormalities were observed in most participants and were independently predicted by hypertension, affirming the role of TTE for CVD screening in this population aged older than 50 years.

Reactive strength index as a key performance indicator in different athlete populations – A systematic review. Science & Sports, 39(2): 129-143, 2024.

Objectives: Reactive Strength Index (RSI) is a neuromuscular assessment used to measure stretch–shortening cycle (SSC) function of the lower limb. Increased RSI scores have been frequently associated with running performance, such as faster sprint times, greater running economy and reduced change of direction times. These locomotor activities are involved in various sports and therefore, RSI is an important indicator of athletic capacity in sport. Whilst RSI has been commonly used in studies to assess SSC performance, no review has been conducted examining the difference in RSI between athletic populations. Thus, the aim of this systematic review was to summarize the literature and determine whether there was any significant difference in RSI between different populations and summarize the underlying factors contributing to these results.

Methods: The systematic review protocol yielded 865 articles for screening by which 19 were included for final analysis following inclusion/exclusion criteria.

Results: Moderate to large differences were predominately found between different athlete populations, and the strength of these relationships varied depending on the populations being compared.

Conclusion: The findings indicated that relative strength is an influencing factor that differentiated RSI performance between populations. This should be an area, which practitioners aim to develop if wanting to improve reactive strength.

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Relationship between athletes' history of stressors and sport injury: A systematic review and meta-analysis. Perceptual and Motor Skills, 131(1): 192-218, 2024.

Abstract - A history of stressors in athletes represents psychosocial factors that may lead to sport injury. However, empirical studies have provided varying results for the relationship between stress history and sport injury. We examined prior literature on the stress history - sport injury relationship within a systematic review and, by meta-analysis, we offered a pooled estimate of the strength of this relationship. We searched seven major academic databases (Sportdiscus, Psyinfo, Academic Search Premier, Ovid, Scopus, Web of Science, and PubMed) from January 2000 to September 2023 and identified 19 empirical studies that examined injuries in sports contexts for meta-analysis. In 19 empirical studies of moderate to high publication quality, we found moderate heterogeneity ($Q(17) = 98.61$; $p < .001$), low sensitivity (I2 77.82-83.77), and low publication bias (Z-value = 7.74; $p < .001$). Further, using a random effect estimate-r, we found a low but significant correlation between stress history and sport injury, yielding a small overall effect size (ES) of $r = .12$. Furthermore, moderation analyses found adolescents ($r = .14$), contact-sport athletes ($r = .09$), non-elite athletes ($r = .13$), and non-European athletes (America $r = .16$; Asia $r = .14$; Oceania $r = .14$) to have a relatively higher ES than their counterparts in this stress history/sport injury relationship. We concluded that inevitable life stressors may lead to many negative consequences for athletes, such that sports professionals should provide stress management educational programs to enhance athletes' health and well-being.

Reliability and validity of the end range hamstring strength test with handheld dynamometry. International Journal of Sports Physical Therapy, 19(3): 268-274, 2024.

Background and Purpose: Hamstring strength testing is crucial in assessing this important muscle group during rehabilitation and injury prevention. The purpose of this study was to evaluate the reliability and validity of measuring strength in a maximally lengthened position using a handheld dynamometer (HHD) compared to isokinetic testing.

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Methods: Twenty healthy and active participants were recruited, and isometric strength testing was performed both on the Biodex machine and isometric end-range hamstring testing with the HHD on two occasions.

Results: Significant strong correlations ($r = 0.87$, $p < 0.0001$) were found between strength testing on the Biodex machine and the HHD for both the right and left sides. Bland-Altman plots demonstrated agreement between the two measurements. Excellent intra-rater reliability (ICC= 0.91-0.93) and moderate to good inter-rater reliability (ICC = 0.71-0.83) for the HHD test was demonstrated.

Conclusions: The HHD provides a reliable and valid alternative to isokinetic testing for assessing hamstring strength in a maximally lengthened position. This assessment can be valuable in monitoring rehabilitation progress and detecting deficits before and after returning to play. The HHD is an accessible and cost-effective option for assessing hamstring strength at long lengths in a clinical setting.

Level of evidence: 3

Responses from ChatGPT-4 show limited correlation with expert consensus statement on anterior shoulder instability. Arthroscopy, Sports Medicine, and Rehabilitation, 100923, 2024.

Purpose: To compare the similarity of answers provided by Generative Pretrained Transformer-4 (GPT-4) with those of a consensus statement on diagnosis, nonoperative management, and Bankart repair in anterior shoulder instability (ASI).

Methods: An expert consensus statement on ASI published by Hurley et al. in 2022 was reviewed and questions laid out to the expert panel were extracted. GPT-4, the subscription version of ChatGPT, was queried using the same set of questions. Answers provided by GPT-4 were compared with those of the

expert panel and subjectively rated for similarity by 2 experienced shoulder surgeons. GPT-4 was then used to rate the similarity of its own responses to the consensus statement, classifying them as low, medium, or high. Rates of similarity as classified by the shoulder surgeons and GPT-4 were then compared and interobserver reliability calculated using weighted k scores.

Results: The degree of similarity between responses of GPT-4 and the ASI consensus statement, as defined by shoulder surgeons, was high in 25.8%, medium in 45.2%, and low 29% of questions. GPT-4 assessed similarity as high in 48.3%, medium in 41.9%, and low 9.7% of questions. Surgeons and GPT-4 reached consensus on the classification of 18 questions (58.1%) and disagreement on 13 questions (41.9%).

Conclusions: The responses generated by artificial intelligence exhibit limited correlation with an expert statement on the diagnosis and treatment of ASI. Clinical Relevance: As the use of artificial intelligence becomes more prevalent, it is important to understand how closely information resembles content produced by human authors.

Routine, Routine, Routine: Sleep regularity and its association with sleep metrics in professional Rugby Union athletes. Sports Medicine – Open, 10: 151, 2024.

Background: Maintaining a consistent sleep and wake time is often reported as a key component of circadian rhythmicity and quality sleep. However, the impact of sleep onset and offset time variability on overall sleep outcomes are underreported in elite athlete populations. This study investigated the relationship between sleep onset and offset time variability using the sleep regularity index (SRI) and measures of sleep and well-being in professional rugby union athletes. Twenty-three professional male rugby union athletes (mean ± SD, age: 23 ± 3 y) underwent sleep monitoring via wrist actigraphy for three weeks during a pre-season phase of training and completed a daily wellness questionnaire. Median SRI was calculated and used to stratify the trainees into two quantile groups: >76.4 SRI (Regular, n = 11) and < 76.4 SRI (Irregular, n = 12).

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Results: The regular sleep group showed significantly longer total sleep duration ($p = 0.02$, $d = 0.97$) compared to the irregular group (7:42 ± 0:29 vs. 7:18 ± 0:20 h: min per night, respectively). Furthermore, while not statistically significant, the regular sleep group showed greater sleep efficiency and less wake episodes compared to irregular sleepers, as demonstrated by moderate effect sizes ($d = 0.71$ and 0.69, respectively).

Conclusions: The results from this study indicate that minimizing variability in sleep onset and offset time is beneficial for increasing sleep duration and may improve sleep efficiency during pre-season training in elite male rugby union athletes. This study provides evidence for the importance of including sleep-wake routines as a key component of sleep education interventions.

Key Points

•When compared to irregular sleepers, regular sleepers demonstrated significantly longer total sleep duration (~24 min per night) over a three-week period (>9 h more of total sleep).

•Regular sleepers experienced fewer wake episode during the night, and had moderately higher sleep efficiency than irregular sleepers.

•The median sleep regularity index in this male, professional rugby cohort was 76.4, considerably lower than that reported in previous reports of elite team sport athletes.

•This study provides further support for the inclusion of promoting regular sleep-wake routines as a key component of sleep hygiene education.

Serum myokines as potential biomarkers of myostatin inhibition in sport doping: A preliminary study on their baseline levels in elite athletes. Biology of Sport, 41(2): 175-183, 2024.

Abstract - We considered in this study the possibility of developing an indirect procedure for detecting myostatin inhibition/suppression, a practice that is prohibited as doping in sport. We have specifically considered the potential diagnostic utility of human serum myokines as indirect markers of myostatin inhibition. Myostatin, its main antagonist follistatin, and other myokines (follistatin-like 1, musclin,

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oncostatin, osteonectin, irisin, brain derived neurotrophic factor, and insulin-like growth factor-1) were selected as a panel of potential biomarkers whose levels may be altered following myostatine suppression. The serum levels of myostatin and of the nine myokines were measured in elite athletes of different age, sex, and sport discipline, and their cross correlation assessed by multivariate analysis. All myokines resulted to be measurable in human serum, except for musclin and irisine, whose levels were below the limits of quantitation in a reduced number of samples. Serum concentrations varied of different orders in magnitude (musclin and osteonectin < 1 ng/mL; follistatin, myostatine and irisine 1-5 ng/mL; brainderived neurotrophic factor, follistatin-like 1 and iinsulin-like growth factor-1 > 10 ng/mL), while no significant differences were found between female and male subjects, with the exceptions of follistatin-like 1 and musclin, showing a higher concentration in females ($p < 0.05$). Levels of insulin-like growth factor 1 and brain derived neurotrophic factor were significantly higher in power athletes than in endurance ones. Multivariate statistics showed that musclin, follistatin-like 1 and oncostatin are more clustered and correlated to myostatin than other myokines, suggesting they could be considered as potential biomarkers of doping by myostatin inhibitors.

Team Physician Consensus Statement: Return to Sport/Return to Play and the Team Physician: A Team Physician Consensus Statement – 2023 Update. Medicine and Science in Sports and Exercise, 56(5): 767-775, 2024.

Overview: Return to play (RTP) is the process of returning an athlete to participate in his/her/their sport. Return to sport (RTS) is applicable to all sports and athletes. For the purposes of this consensus statement, RTS will be used to include both the process and the decision, focused on non-game-day RTS. It is important for the team physician to recognize RTS represents a continuum: return to participation, RTS, and return to performance. This progression can be applied for any sport, athlete, or injury/illness. The RTS decision is ongoing, is context dependent, and may change over time. The team physician has the central role in not only protecting the athlete's health, but also protecting the athlete from coercion to participate. The final RTS decision should be led by the team physician as part of a shared decision-making process (SDM). This model remains the best practice in making the RTS decision. The process is dynamic, and decision points may change over time based on evolving information and risk. SDM is a

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model of patient-centered care that enables and encourages patients to participate in medical decisions that affect their health. It operates under two premises:

• Patients empowered with information will participate in the medical decision-making process by asking informed questions and expressing personal values and opinions about their conditions and treatment options.

• Physicians respect patients' goals and preferences and use them to guide recommendations and treatments.

For the team physician, an athlete plays a central role in SDM, providing input and direction for the RTS decision. Other stakeholders may be involved in the process. SDM goals achieve athlete-centered care with these objectives:

• Present the science and unknowns regarding the diagnosis

• Summarize individual athlete risk profiles

• Discuss evidence-based options and plans considering the athlete's values and preferences

Injury and illness are common in sport, and RTS decisions are made using a variety of established frameworks and criteria. The decision requires understanding the nature of the injury/illness and athlete- and sport-specific factors in the context of SDM. Challenges include the level of existing and evolving science and the lack of consensus in the clinical community. Age, sex, sport, time of season, athlete risk tolerance, and psychological readiness are other factors to consider in the RTS process. Physician and organizational risk tolerance and medicolegal issues should also be considered, along with existing policies and consensus recommendations.

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The acute:chronic workload ratio and injury risk in semiprofessional football players. International Journal of Sports Medicine, Ahead of Print: 1-25, 2024.

Abstract - The purpose of this study was to analyze the association and predictive capacity between the acute:chronic workload ratio (ACWR) and non-contact injuries in a semiprofessional football team. Seventeen football or soccer players from a Spanish Third Division football team participated voluntarily in this study. A prospective longitudinal study was developed during the 2020/2021 season. Twenty-four weeks were analyzed from October to March, including a regenerative microcycle due to the absence of competition during Christmas. Rate of perceived exertion (RPE) and session-rate of perceived exertion (sRPE) were registered for every training and game session. Afterward, acute and chronic workloads were calculated, and ACWR was subsequently derived from them. Furthermore, non-contact injuries were registered during the period mentioned. The main findings were that there is a poor correlation between the ACWR and non-contact injuries ($r=0.069$ ($p<0.05$)), and the use of the ACWR by itself is insufficient to predict the occurrence of non-contact injuries in a semiprofessional football team. Consequently, the ACWR is not an useful predictive tool for injuries in semiprofessional football teams.

The fundamentals and applications of wearable sensor devices in sports medicine: A scoping review. Arthroscopy: The Journal of Arthroscopic and Related Surgery, Ahead of Print: 1-20, 2024.

Purpose: To (1) characterize the various forms of wearable sensor devices (WSDs) and (2) review the peer-reviewed literature of applied wearable technology within sports medicine.

Methods: A systematic search of PubMed and EMBASE databases, from inception through 2023, was conducted to identify eligible studies using WSDs within sports medicine. Data extraction was performed of study demographics and sensor specifications. Included studies were categorized by application: athletic training, rehabilitation, and research.

Results: In total, 43 studies met criteria for inclusion in this review. Forms of WSDs include pedometers, accelerometers, encoders (consisting of magnetometers and gyroscopes), force sensors, global positioning system trackers, and inertial measurement units. Outcome metrics include step counts; gait,

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limb motion, and angular positioning; foot and skin pressure; change of direction and inclination, including analysis of both body parts and athletes on a field; displacement and velocity of body segments and joints; heart rate; plethysmography; sport-specific kinematics; range of motion, symmetry, and alignment; head impact; sleep; throwing biomechanics; and kinetic and spatiotemporal running metrics. WSDs are used in athletic training to assess sport-specific biomechanics and workload with a goal of injury prevention and training optimization, as well as for rehabilitation monitoring and research such as for risk predicting and aiding diagnosis.

Conclusions: WSDs enable real-time monitoring of human performance across a variety of implementations and settings, allowing collection of metrics otherwise not achievable. WSDs are powerful tools with multiple applications within athletic training, patient rehabilitation, and orthopaedic and sports medicine research.

Clinical Relevance: Wearable technology may represent the missing link to quantitatively addressing return to play and previous performance. WSDs are commercially available and portable adjuncts that allow clinicians, trainers, and individual athletes to monitor biomechanical parameters, workload, and recovery status to better contextualize personalized training, injury risk, and rehabilitation.

Editorial: The hidden cardiovascular crisis among former NFL athletes. Journal of the American College of Cardiology, 83(19): 1838-1840, 2024. (No abstract available)

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The Maximal Intensity Period: Rationalizing its use in team sports practice. Sports Medicine – Open, 8:128, 2022.

Abstract - Quantifying the highest intensity of competition (the maximal intensity period [MIP]) for varying durations in team sports has been used to identify training targets to inform the preparation of players. However, its usefulness has recently been questioned since it may still underestimate the training intensity required to produce specific physiological adaptations. Within this conceptual review, we aimed to: (i) describe the methods used to determine the MIP; (ii) compare the data obtained using MIP or whole-match analysis, considering the influence of different contextual factors; (iii) rationalize the use of the MIP in team sports practice and (iv) provide limitations and future directions in the area. Different methods are used to determine the MIP, with MIP values far greater than those derived from averaging across the whole match, although they could be affected by contextual factors that should be considered in practice. Additionally, while the MIP might be utilized during sport-specific drills, it is inappropriate to inform the intensity of interval-based, repeated sprint and linear speed training modes. Lastly, MIP does not consider any variable of internal load, a major limitation when informing training practice. In conclusion, practitioners should be aware of the potential use or misuse of the MIP.

Key Points

• Determining the maximal intensity period can inform the training of the most intense demands in team sports matches

• Maximal intensity period can be used as a bench mark for sport-specific drills such as small-sided games

• Maximal intensity period underestimates the actual player's physical capacities and should not be used as a benchmark in running-based exercises

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The prevalence and influence of new or worsened neck pain after a sport-related concussion in collegiate athletes: A study from the CARE Consortium. The American Journal of Sports Medicine, Ahead of Print: 1-10, 2024.

Background: Neck pain in a concussion population is an emerging area of study that has been shown to have a negative influence on recovery. This effect has not yet been studied in collegiate athletes.

Hypothesis: New or worsened neck pain is common after a concussion (>30%), negatively influences recovery, and is associated with patient sex and level of contact in sport.

Study design: Cohort study; Level of evidence, 2.

Methods: Varsity-level athletes from 29 National Collegiate Athletic Association member institutions as well as nonvarsity sport athletes at military service academies were eligible for enrollment. Participants completed a preseason baseline assessment and follow-up assessments at 6 and 24 to 48 hours after a concussion, when they were symptom-free, and when they returned to unrestricted play. Data collection occurred between January 2014 and September 2018.

Results: A total of 2163 injuries were studied. New or worsened neck pain was reported with 47.0% of injuries. New or worsened neck pain was associated with patient sex (higher in female athletes), an altered mental status after the injury, the mechanism of injury, and what the athlete collided with. The presence of new/worsened neck pain was associated with delayed recovery. Those with new or worsened neck pain had 11.1 days of symptoms versus 8.8 days in those without ($P < .001$). They were also less likely to have a resolution of self-reported symptoms in ≤7 days ($P < .001$). However, the mean duration of the return-to-play protocol was not significantly different for those with new or worsened neck pain (7.5 ± 7.7 days) than those without (7.4 ± 8.3 days) ($P = .592$).

Conclusion: This novel study shows that neck pain was common in collegiate athletes sustaining a concussion, was influenced by many factors, and negatively affected recovery.

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The relationship between isometric and dynamic strength following resistance training: A systematic review, meta-analysis, and level of agreement. International Journal of Sports Physiology and Performance, 19(1): 2-12, 2023.

Background: Maximal lower-body strength can be assessed both dynamically and isometrically; however, the relationship between the changes in these 2 forms of strength following resistance training is not well understood.

Purpose: To systematically review and analyze the effects of resistance training on changes in maximal dynamic (1-repetition-maximum back squat, deadlift, and power clean) and position-matched isometric strength (isometric midthigh pull and the isometric squat). In addition, individual-level data were used to quantify the agreement and relationship between changes in dynamic and isometric strength.

Methods: Databases were systematically searched to identify eligible articles, and meta-analysis procedures were performed on the extracted data. The raw results from 4 studies were acquired, enabling bias and absolute reliability measures to be calculated using Bland-Altman test of agreement.

Results: Eleven studies met the inclusion criteria, which resulted in 29 isometric-dynamic change comparisons. The overall pooled effect was 0.13 in favor of dynamic testing; however, the prediction interval ranged from g = -0.49 to 0.75. There was no evidence of bias (P = .825) between isometric and dynamic tests; however, the reliability coefficient was estimated to be 16%, and the coefficient of variation (%) was 109.27.

Conclusions: As a range of future effects can be expected when comparing isometric to dynamic strength changes following resistance training, and limited proportionality exists between changes in these 2 strength qualities, there is strong evidence that isometric and dynamic strength represent separate neuromuscular domains. These findings can be used to inform strength-assessment models in athlete populations.

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